PROBLEM VF14

Arm Approaching a Wall

Original statement · VF14.pdf

Squeeze film | Closed end | Force and torque

OFFICIAL PROBLEM STATEMENT

The L-shaped arm of length L depicted in the figure moves vertically approaching the horizontal wall, forming a thin channel of decreasing uniform thickness $h(t) \ll L$. A liquid of density ρ and viscosity μ fills the channel. Due to the arm displacement, the liquid is forced to move out, discharging to the surrounding atmosphere, where the reduced pressure is P_∞. For a known value of $h(t)$,

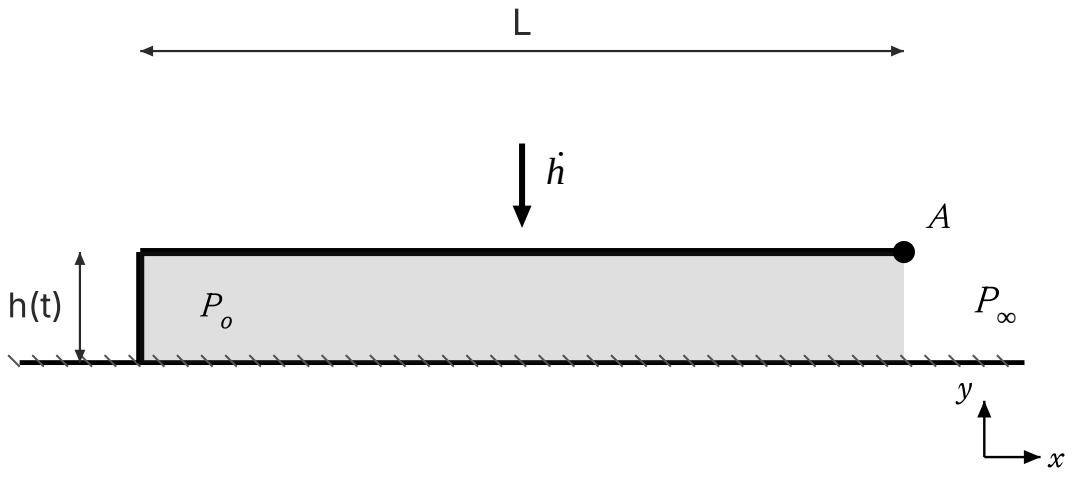
1. Give the conditions for the motion in the channel to be dominated by viscosity.

2. Determine the velocity profile across the channel $v_x(x, y, t)$, the associated volume flux $Q = \int_0^h v_x \, dy$, and the reduced pressure distribution $P(x, t)$, including the value $P_0(t)$ at $x = 0$.

3. Calculate the force acting on the arm $\bar{F} = (F_x, F_y)$.

4. Obtain the torque exerted on the arm with respect to its end point A.

The sheet writes the pressure at $x = 0$ as P_o in the figure and P_0 in the statement.



Statement figure (redrawn from VF14.pdf): an L-shaped arm whose horizontal bar of length L is at the height h(t) above a wall and whose vertical leg closes the channel at x = 0; A is the end of the bar and P_∞ the pressure outside; P_o is the pressure at x = 0

Source: VF14.pdf (page 1, exam of Jan 10, 2012, statement and handwritten key). Notes.pdf, Chapter 7: slender films, Eqs. 7.67-7.71; the wedge effect, Eq. 7.72; forces and moments, Eq. 7.76.

Worked solution

Hypotheses, Data and Unknowns

- The arm moves vertically with the velocity $\dot{h} = dh/dt$ (negative when it approaches the wall); the channel, $0 < x < L, 0 < y < h(t)$, is closed at $x = 0$ by the vertical leg and open at $x = L$, where the pressure is P_∞.
- Slender film ($h \ll L$), pressure uniform across it; gravity is absorbed in the reduced pressure.

Criterion for Viscous Dominance

By continuity $V_L \sim L\dot{h}/h$ and $t_0 \sim h/|\dot{h}|$. The convective and local accelerations against the viscous force (Eqs. 7.67-7.68):

$$\frac{\rho V_L h}{\mu}\frac{h}{L} \sim \frac{L\dot{h}}{h}\frac{h}{\nu}\frac{h}{L} = \frac{h\dot{h}}{\nu} \ll 1, \qquad \frac{h^2/\nu}{t_0} \sim \frac{h\dot{h}}{\nu} \ll 1$$

So $h\,|\dot{h}|/\nu \ll 1$.

Velocity, Flux and Pressure

Equation (7.69) with no slip on the wall and on the arm ($v_x = 0$ at $y = 0$ and $y = h$) gives the Poiseuille profile (7.11):

$$0 = P_l + \mu\frac{\partial^2 v_x}{\partial y^2} \implies v_x = \frac{P_l}{2\mu}\,y\,(h - y), \qquad Q = \frac{P_l h^3}{12\mu} = -\frac{\partial P}{\partial x}\frac{h^3}{12\mu}$$

Integrating the continuity equation across the channel, with $v_x(h) = 0$, $v_y(0) = 0$ and $v_y(h) = \dot{h}$ (the arm moves with $\dot{h}$):

$$\int_0^h \left(\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y}\right) dy = 0 \implies \frac{\partial Q}{\partial x} + \dot{h} = 0$$

so $Q = -\dot{h}x$ (zero at the closed end $x = 0$; positive for an approaching arm). Then $-\frac{\partial P}{\partial x}\frac{h^3}{12\mu} = -\dot{h}x$, that is, the first integration of the key, $\frac{\partial P}{\partial x}\frac{h^3}{12\mu} = \dot{h}x$ (the key's boxed line carries a sign that is the opposite of its next line, which is the one used):

$$\frac{\partial P}{\partial x} = \frac{12\mu\dot{h}}{h^3}\,x \implies P = \frac{6\mu\dot{h}}{h^3}\,x^2 + C_2$$

With $P(L) = P_\infty$:

$$P(x,t) = P_\infty + \frac{6\mu\dot{h}}{h^3}\left(x^2 - L^2\right), \qquad P_0(t) = P_\infty - \frac{6\mu\dot{h}L^2}{h^3}$$

For an approaching arm ($\dot{h} < 0$) there is an overpressure, largest at the closed end.

Force on the Arm

Vertical force. The liquid below the arm pushes it upward with the excess pressure (the pressure P_∞ acts on the whole surface of the arm and gives no net force):

$$F_y = \int_0^L (P - P_\infty)\,dx = \frac{6\mu\dot{h}}{h^3}\left[\frac{x^3}{3} - L^2 x\right]_0^L = \frac{6\mu\dot{h}}{h^3}\left(-\frac{2L^3}{3}\right) = -\frac{4\mu\dot{h}L^3}{h^3}$$

positive for $\dot{h} < 0$: the liquid resists the approach (it is a force of the order of $\mu|\dot{h}|L^3/h^3$, very large for a thin film).

Horizontal force. Two contributions: the pressure on the inner face of the vertical leg (area h, normal $+\bar{e}_x$ from the arm into the liquid; the force of the liquid is $-(P_0 - P_\infty)h$) and the shear on the underside of the arm ($\bar{n} = -\bar{e}_y$, force $-\mu\,\partial v_x/\partial y$ per unit area):

$$F_x = -(P_0 - P_\infty)\, h - \int_0^L \mu \left.\frac{\partial v_x}{\partial y}\right|_{y=h} dx, \qquad -(P_0 - P_\infty)h = \frac{6\mu \dot{h} L^2}{h^2}$$

With $\partial v_x/\partial y = \frac{P_l}{2\mu}(h - 2y)$, $P_l = -\partial P/\partial x = -12\mu\dot{h}x/h^3$: at $y = h$, $\mu\,\partial v_x/\partial y = -\frac{P_l h}{2} = \frac{6\mu\dot{h}x}{h^2}$, and

$$-\int_0^L \frac{6\mu\dot{h}x}{h^2}\, dx = -\frac{3\mu\dot{h}L^2}{h^2}, \qquad F_x = \frac{6\mu\dot{h}L^2}{h^2} - \frac{3\mu\dot{h}L^2}{h^2} = \frac{3\mu\dot{h}L^2}{h^2}$$

$$\bar{F} = \left(\frac{3\mu\dot{h}L^2}{h^2},\ -\frac{4\mu\dot{h}L^3}{h^3}\right), \qquad |F_x|/|F_y| \sim h/L \ll 1$$

Torque about the End Point A

The pressure on the underside of the arm gives the moment about A ($x = L$) of a vertical force at the distance $x - L$ (arm positive along $\bar{e}_x \times \bar{e}_y = \bar{e}_z$):

$$\bar{M} = -\int_0^L (P - P_\infty)(L - x)\, dx\ \bar{e}_z = -\frac{6\mu\dot{h}}{h^3}\int_0^L \left(x^2 - L^2\right)(L - x)\, dx\ \bar{e}_z$$

$$\int_0^L \left(x^2 - L^2\right)(L - x)\, dx = \int_0^L \left(Lx^2 - x^3 - L^3 + L^2 x\right) dx = L\frac{L^3}{3} - \frac{L^4}{4} - L^4 + \frac{L^4}{2} = L^4\left(\frac{1}{3} - \frac{1}{4} - 1 + \frac{1}{2}\right) = -\frac{5}{12}L^4$$

$$\bar{M} = \frac{5}{2}\,\frac{\mu\dot{h}}{h^3}\,L^4\ \bar{e}_z$$

The moment of the pressure on the leg is of order $\mu\dot{h}L^2/h$ (the pressure $\sim \mu\dot{h}L^2/h^3$ acting on the area h with the arm $\sim h$) and the moment of the shear on the arm is $\sim \mu\dot{h}L^2/h$: a factor $(h/L)^2$ smaller than the one above and neglected, as the key does.

Answers and Checks

- Viscous dominance: $h|\dot{h}|/\nu \ll 1$.
- Profile and flux: $v_x = \frac{P_l}{2\mu} y(h-y), Q = -\dot{h}x$; pressure: $P = P_\infty + \frac{6\mu\dot{h}}{h^3}(x^2 - L^2)$, $P_0 = P_\infty - \frac{6\mu\dot{h}L^2}{h^3}$.
- Force: $\bar{F} = \left(\frac{3\mu\dot{h}L^2}{h^2}, -\frac{4\mu\dot{h}L^3}{h^3}\right)$; torque about A: $\bar{M} = \frac{5}{2}\frac{\mu\dot{h}L^4}{h^3}\bar{e}_z$.
- Dimensions: $\mu\dot{h}L^3/h^3$ is $ML^{-1}T^{-1} \cdot LT^{-1} \cdot L^3/L^3 = MLT^{-2} \cdot L^{-1}$, a force per unit width.
- Limits: $\dot{h} \to 0$ gives zero force, torque and overpressure; $h \to 0$ makes them grow as h^{-3} (and the film does not drain in finite time at a constant $\dot{h}$ without infinite force).
- Check against the official key: VF14.pdf gives the same criterion, profile, flux, pressure, force and torque; its boxed continuity line has the opposite sign of the next one (only the second is consistent with $\partial Q/\partial x = -\dot{h}$).